Chapter 5. The Tychonoff Theorem

Compactness of products, Stone–Čech compactification

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5.1 The Tychonoff theorem

Theorem 5.1 (Tychonoff)

An arbitrary product of compact spaces is compact in the product topology.

We use the finite intersection property (FIP) characterization of compactness.

Lemma 5.2

Every collection $\mathcal{A}$ of subsets of X with the FIP is contained in a collection $\mathcal{D}$ that is maximal with respect to the FIP. Such a $\mathcal{D}$ is closed under finite intersections, and any set meeting every member of $\mathcal{D}$ belongs to $\mathcal{D}$.

Proof. Zorn's lemma applied to the collections containing $\mathcal{A}$ with the FIP, ordered by inclusion (the union of a chain still has the FIP, since any finite subfamily lies in one member of the chain). If A meets every member of $\mathcal{D}$, then $\mathcal{D} \cup \{\text{finite intersections of } \mathcal{D}\} \cup \{A\}$ still has the FIP, so by maximality $A \in \mathcal{D}$. ■

Proof of Tychonoff's theorem. Let $\mathcal{A}$ be a collection of subsets of $X = \prod X_\alpha$ with the FIP; we show $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$. Enlarge $\mathcal{A}$ to a maximal $\mathcal{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so pick $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$ and let $x = (x_\alpha)$. For any subbasis set $\pi_\beta^{-1}(U_\beta)$ containing x, U_β meets every $\pi_\beta(D)$, so $\pi_\beta^{-1}(U_\beta)$ meets every D and therefore belongs to $\mathcal{D}$. Finite intersections of these (the basis sets around x) also lie in $\mathcal{D}$ and hence meet every D. So $x \in \overline{D}$ for all $D \in \mathcal{D} \supset \mathcal{A}$. ■

The theorem is equivalent to the axiom of choice. It fails for the box topology, and for infinite products only the product topology is compact (e.g. $[0,1]^\omega$ is compact, $\mathbb{R}^\omega$ is not since $\mathbb{R}$ is not).

Example 5.1

(a) The Hilbert cube $[0,1]^\omega$ is a compact metrizable space, and by Urysohn's theorem every second countable regular space imbeds in it.
(b) $\{0,1\}^\omega$ is compact and homeomorphic to the Cantor set; $[0,1]^J$ for uncountable J is compact but neither metrizable nor sequentially compact.

5.2 The Stone–Čech compactification

A **compactification** of X is a compact Hausdorff Y containing X as a dense subspace. Only completely regular spaces have compactifications, and every completely regular space has them: imbed X in a cube $[0,1]^J$ and take the closure.

Theorem 5.3

Let X be completely regular. There is a compactification βX such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous function on βX. More generally every continuous map f of X into a compact Hausdorff space C extends uniquely to $\beta X \to C$. βX is unique up to a homeomorphism fixing X.

Proof. Let $\{f_\alpha\}_{\alpha\in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset [a_\alpha, b_\alpha]$. $h = (f_\alpha) : X \to \prod[a_\alpha, b_\alpha]$ is an imbedding (complete regularity), and $\beta X = \overline{h(X)}$ is compact by Tychonoff. Each f_α extends as the projection π_α. For a compact Hausdorff C, imbed C in a cube $[0,1]^K$; the coordinates of f extend, the extension maps $\beta X = \overline{X}$ into $\overline{f(X)} \subset C$ since C is closed.

Uniqueness: extensions agreeing on a dense set agree (Hausdorff target), and the universal property identifies any two such compactifications. ■

Example 5.2

(a) The one-point compactification of $(0,1)$ is a circle; $[0,1]$ is a two-point compactification; $\beta(0,1)$ is much larger, since $\sin(1/x)$ extends.
(b) $\beta\mathbb{Z}_+$ has the cardinality of $\mathcal{P}(\mathcal{P}(\mathbb{Z}_+))$ and is not sequentially compact; it can be described as the space of ultrafilters on $\mathbb{Z}_+$.
(c) $\beta S_\Omega = \overline{S}_\Omega$: every continuous real function on S_Ω is eventually constant.